MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Predicting Life Insurance Lapses: Integrating Cox Proportional Hazards and Random Survival Forest

By

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Chapter 1

Introduction

1.1 Background

Life insurance is a cornerstone of long-term financial security for households and a stabilizing force for insurers and the broader financial system. A major challenge in the industry is policy lapse, where policyholders discontinue premium payments prematurely. Globally, high lapse rates reduce insurer profitability, disrupt asset-liability management, and increase regulatory capital requirements under frameworks such as Solvency II and IFRS 17 (Gottlieb and Smetters, 2021). Beyond financial metrics, lapses undermine consumer trust and limit the contribution of life insurance to financial inclusion and social protection.

In developing economies, particularly in Sub-Saharan Africa, lapse rates are even more pronounced due to economic volatility, low insurance penetration, and limited financial literacy. In Malawi, the insurance sector is growing but remains underdeveloped, with penetration levels below 5%. Policy lapses exacerbate these challenges by eroding the risk pool, reducing long-term savings mobilization, and constraining market growth. Addressing lapses is therefore critical not only for insurer profitability but also for enhancing household resilience and supporting broader economic development in the region.

To predict lapses, traditional survival analysis methods such as the Cox Proportional Hazards (Cox PH) model have been widely used. Cox PH models offer interpretability by identifying important risk factors such as age, premium levels, and health status (Manteigas and António, 2024). However, their reliance on proportional hazards and linearity assumptions limits their performance when relationships are complex or non-linear. Machine learning approaches, particularly Random Survival Forests (RSF), can capture these non-linear interactions and handle high-dimensional data (Fier and Liebenberg, 2013), but they often lack interpretability, making it challenging for insurers and regulators to translate results into practical retention strategies.

The gap exists in the need for methods that combine predictive accuracy with interpretability, especially in emerging markets like Malawi, where data quality and operational constraints present additional challenges. This study proposes a hybrid

Cox-RSF model that leverages the interpretability of Cox PH and the predictive strength of RSF. By integrating these approaches, the model aims to improve the accuracy of lapse predictions, inform more effective retention strategies, and enhance decision-making in the life insurance sector.

1.2 Problem Statement

Life insurance companies continue to face high policy lapse rates, which reduce profitability and complicate financial planning and solvency management. Policy lapses disrupt asset-liability matching, affect customer retention, and increase capital requirements under frameworks such as IFRS 17 and Solvency II (Gottlieb and Smetters, 2021). Accurate prediction of lapses is therefore essential for maintaining financial stability and designing targeted retention strategies.

Traditional survival analysis methods such as the Cox Proportional Hazards (Cox PH) model have been widely used to identify lapse determinants including age, premium level, and payment method. However, these models assume proportional hazards and linear relationships, which limit their ability to capture complex, non-linear interactions in high-dimensional policyholder data (Reck et al., 2023; Andrade and Valencia, 2021). Machine-learning techniques like the Random Survival Forest (RSF) can address non-linearity and variable interactions effectively, yet they often lack interpretability required for actuarial and regulatory transparency (Andrade and Valencia, 2021; Fier and Liebenberg, 2013).

This methodological limitation creates a gap between models that are interpretable and those that are highly predictive. A hybrid Cox-RSF model offers a potential solution by combining the interpretability of the Cox PH framework with the predictive strength of RSF. Developing such a hybrid model can improve the accuracy and practical usefulness of lapse prediction, helping insurers to anticipate risk and implement data-driven retention strategies (Wang et al., 2023; Gupta et al., 2025).

1.3 Research Objectives

Main Research Objective

- To develop and evaluate a hybrid Cox Proportional Hazards-Random Survival Forest (Cox-RSF) model for accurate and interpretable prediction of life insurance policy lapses.

Specific Research Objectives

- To construct a hybrid Cox-RSF model that integrates interpretability from Cox PH and predictive strength from RSF.
- To apply the developed hybrid model to predict the time to lapse of life insurance policies using real or secondary data.
- To compare the performance of the hybrid model with standalone Cox PH and RSF models using appropriate evaluation metrics.

1.4 Research Questions

1. How can the hybrid Cox-RSF model be developed to integrate the interpretability of Cox PH with the predictive strength of RSF?
2. To what extent does the hybrid Cox-RSF model improve lapse prediction accuracy compared to standalone Cox PH and RSF models?
3. How well does the hybrid model capture non-linear effects and complex interactions influencing policy lapse behavior?

1.5 Purpose of the Study

The purpose of this study is to develop and evaluate a hybrid Cox Proportional Hazards-Random Survival Forest (Cox-RSF) model to improve the prediction of life insurance policy lapses. Academically, the study contributes to actuarial science and predictive modeling by integrating survival analysis and machine learning algorithms, addressing the limitations of traditional methods such as Cox PH and RSF. Practically, it provides insurers with a more accurate and interpretable tool for identifying lapse risk, designing effective retention strategies, and enhancing financial stability under regulatory frameworks such as IFRS 17 and Solvency II. Additionally, the findings are expected to inform better policyholder management practices within the Malawian life insurance sector, supporting growth, customer trust, and overall market development.

Chapter 2

Literature Review

2.0.1 Overview

This literature review examines methods for predicting life insurance policy lapses, focusing on traditional survival analysis and machine learning approaches. Policy lapses-premature terminations due to non-payment or surrender-cost insurers an estimated 5-10% of annual premiums and challenge compliance with regulations like IFRS 17 and Solvency II (Carlos, 2021). The Cox Proportional Hazards (Cox PH) model is widely used to identify lapse drivers such as age, premium levels, payment methods, and health status, providing interpretable hazard ratios (Schumacher and Others, 2025; Andrade and Valencia, 2021). However, its assumptions of proportional hazards and linearity limit its ability to capture complex interactions in high-dimensional policyholder datasets (Schumacher and Others, 2025). Random Survival Forests (RSF) overcome these limitations by modeling non-linear relationships and censored data, achieving higher predictive accuracy (e.g., C-index 0.73) but often lacking interpretability (Andrade and Valencia, 2022, 2021). Hybrid Cox-RSF models, successful in medical survival analysis (e.g., C-index 0.82 for sepsis), combine interpretability with predictive power but remain untested in insurance (Wang et al., 2023). This study addresses this gap by developing a Cox-RSF hybrid to predict lapse risk, evaluating its performance against standalone Cox PH and RSF using metrics like C-index and integrated Brier score (IBS).

2.0.2 Developing a Hybrid Cox-RSF Model for Predicting Policy Lapse

Life insurance policy lapses reduce profitability and complicate liquidity and regulatory compliance under IFRS 17 and Solvency II (Carlos, 2021). Cox PH models effectively identify lapse drivers like age, premiums, and economic factors but are constrained by linearity assumptions, limiting their ability to model complex interactions (Schumacher and Others, 2025; Andrade and Valencia, 2021). RSF, a non-parametric approach, uses log-rank splitting and cumulative hazard estimation to capture non-linear patterns and handle censored data, achieving superior predictive performance in insurance datasets (Andrade and Valencia, 2022, 2021). Hybrid Cox-RSF models, applied in medical contexts, integrate Cox PH's interpretability (e.g., via Elastic Net regularization) with RSF's flexibility, enhancing robustness for censored data (Wang et al., 2023). Existing research on insurance policy lapses has largely applied either traditional survival models or machine learning approaches independently. The integration of Cox PH and RSF for this purpose remains underexplored (Gupta et al., 2025). This study develops such a hybrid model to enhance prediction and retention strategies.

2.0.3 Using the Hybrid Model to Predict Life Insurance Policy Lapse

Accurate lapse prediction enables insurers to design retention strategies, optimize capital allocation, and manage solvency risks (Carlos, 2021). Cox PH models provide interpretable insights into lapse factors but underperform in complex datasets due to linearity constraints (e.g., C-index 0.70 in mortgage-linked insurance) (Schumacher and Others, 2025; Andrade and Valencia, 2021). RSF models excel in capturing non-linear effects and variable interactions, improving predictive accuracy but lacking explainability for regulatory needs (Schumacher and Others, 2025; Andrade and Valencia, 2021). Hybrid Cox-RSF models, successful in medical survival analysis, enhance survival probability estimation and risk stratification across censored datasets (Wang et al., 2023). This study applies a Cox-RSF hybrid to estimate lapse risks at policyholder and portfolio levels, offering a novel framework for actionable predictions.

2.0.4 Significance of the Cox-RSF Hybrid Model in Improving Predictability

Standalone Cox PH and RSF models have distinct limitations: Cox PH offers calibration and interpretability, while RSF captures non-linearities and complex interactions (Schumacher and Others, 2025; Andrade and Valencia, 2022). Integrating these models in a hybrid framework improves predictive performance (e.g., C-index improvements of 0.1 over standalone models) while maintaining explainability for regulatory compliance (Wang et al., 2023). In insurance, such hybrids could reduce lapse-related losses by enabling targeted interventions, such as personalized premium adjustments, and enhance portfolio management (Carlos, 2021; Gupta et al., 2025).

2.0.5 Research Gap

Existing studies on life insurance lapses rely on standalone Cox PH or RSF models, trading off predictive accuracy for interpretability or vice versa (Schumacher and Others, 2025; Andrade and Valencia, 2022, 2021). Cox PH is limited by linearity assumptions, while RSF, despite higher accuracy (e.g., C-index 0.73), lacks interpretability (Schumacher and Others, 2025; Andrade and Valencia, 2022).This study addresses this gap by:

- Developing a Cox-RSF hybrid model for lapse prediction, leveraging Elastic Net-regularized Cox PH and weighted RSF ensembles.
- Applying it to estimate survival probabilities and stratify lapse risks across policyholders and portfolios.
- Evaluating its predictive accuracy (e.g., via C-index and IBS) and interpretability against standalone Cox PH and RSF, using real-world insurance data.

Chapter 3

Methodology

3.1 Overview

This chapter presents the methodological framework for achieving the research objectives. The process involves developing a hybrid Cox-RSF model, applying it to predict life insurance policy lapse, and evaluating its predictive and interpretive performance. The framework integrates statistical and machine learning approaches to ensure interpretability, accuracy, and robustness.

3.2 Developing the Hybrid Cox-RSF Model

The study develops a hybrid model combining outputs from a Cox Proportional Hazards (Cox PH) model and a Random Survival Forest (RSF). This section outlines the model formulation, diagnostic checks, and combination strategy.

Cox Proportional Hazards Model

The Cox PH model will first be fitted to identify key factors influencing policy lapse. The hazard and survival functions are expressed as:

$$h(t|X_i) = h_0(t) \exp(\beta^\top X_i), \tag{3.1}$$

$$S_{Cox}(t|X_i) = \exp\left\{-H_0(t) \exp(\beta^\top X_i)\right\}, \tag{3.2}$$

where:

- $h_0(t)$ = baseline hazard function,
- $H_0(t) = \int_0^t h_0(s)\, ds$ = cumulative baseline hazard,
- $X_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ = covariate vector for policyholder i,
- $\beta = (\beta_1, \beta_2, \ldots, \beta_p)$ = regression coefficients.

The proportional hazards assumption will be tested using **Schoenfeld residuals**. If the assumption is violated, **time-varying effects** or **stratified Cox models** will be applied to account for non-proportional hazards.

Random Survival Forest (RSF)

The RSF model will be trained on the same dataset to capture non-linear relationships and higher-order interactions. Each tree in the forest estimates the cumulative hazard using the Nelson-Aalen estimator:

$$\hat{H}^{(b)}(t|X_i) = \sum_{t_j \le t} \frac{d_j}{n_j}, \tag{3.3}$$

where d_j is the number of lapses at time t_j and n_j is the number of policies at risk just prior to t_j. The survival function for tree b is then:

$$\hat{S}^{(b)}(t|X_i) = \exp\left\{-\hat{H}^{(b)}(t|X_i)\right\}. \tag{3.4}$$

The ensemble survival estimate is obtained as the geometric mean across B trees:

$$\hat{S}_{RSF}(t|X_i) = \left(\prod_{b=1}^{B} \hat{S}^{(b)}(t|X_i)\right)^{1/B}. \tag{3.5}$$

Deriving the Hybrid Model

The hybrid model integrates interpretability from Cox PH and predictive power from RSF using a weighted blending scheme:

$$\hat{S}_{blend}(t|X_i) = w_1\hat{S}_{Cox}(t|X_i) + w_2\hat{S}_{RSF}(t|X_i), \tag{3.6}$$

subject to $w_1 + w_2 = 1$.

The optimal weights (w_1, w_2) will be determined via 5-fold cross-validation by minimizing the Integrated Brier Score (IBS).

3.3 Using the Hybrid Cox-RSF Model to Predict Life Insurance Policy Lapse

The hybrid model will be applied to predict the time to lapse of life insurance policies. The data will be partitioned using **stratified K-fold cross-validation** to ensure balanced event representation across folds. For each policyholder i:

- T_i = observed time to event or censoring,
- $\Delta_i = 1$ if policy i lapsed, 0 if censored.

Missing values will be handled using mean or median imputation for continuous variables, and mode or one-hot encoding for categorical variables. Predicted survival probabilities will be used to rank policyholders by lapse risk:

$$\text{Risk}_i = 1 - \hat{S}_{blend}(t^*|X_i), \tag{3.7}$$

where t^* is a chosen time horizon (e.g., 12 or 24 months).

3.4 Evaluating the Significance of the Hybrid Model

The predictive performance of the hybrid model will be evaluated against standalone Cox PH and RSF models using the following metrics:

Concordance Index (C-index)

An IPCW-adjusted C-index (Harrell's or Uno's) will be used to assess concordance:

$$C = P(\hat{T}_i < \hat{T}_j | T_i < T_j).$$

Time-dependent Brier Score and Integrated Brier Score (IBS)

The time-dependent Brier Score is defined as:

$$BS(t) = \frac{1}{N} \sum_{i=1}^{N} W_i(t) \left(\hat{S}(t|X_i) - Y_i(t) \right)^2, \tag{3.8}$$

where $Y_i(t) = 1$ if policy i has not lapsed by time t, and $W_i(t)$ are IPCW (Inverse Probability of Censoring Weighting) weights. The Integrated Brier Score is:

$$IBS = \frac{1}{T} \int_0^T BS(t)\, dt. \tag{3.9}$$

Calibration and Discrimination

Model calibration will be evaluated using:

- **Calibration slope:** measures agreement between predicted and observed risks.
- **Calibration plots:** graphical assessment of predicted versus observed survival probabilities.

Discrimination will further be assessed using time-dependent ROC-AUC (Receiver Operating Characteristic Curve-Area Under the Curve), capturing model sensitivity and specificity over time.

3.5 Software Environment

The analysis will be implemented in **Python 3.13** (packages: `lifelines`, `scikit-survival`) or **R** (`randomForestSRC`).

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Chapter 4

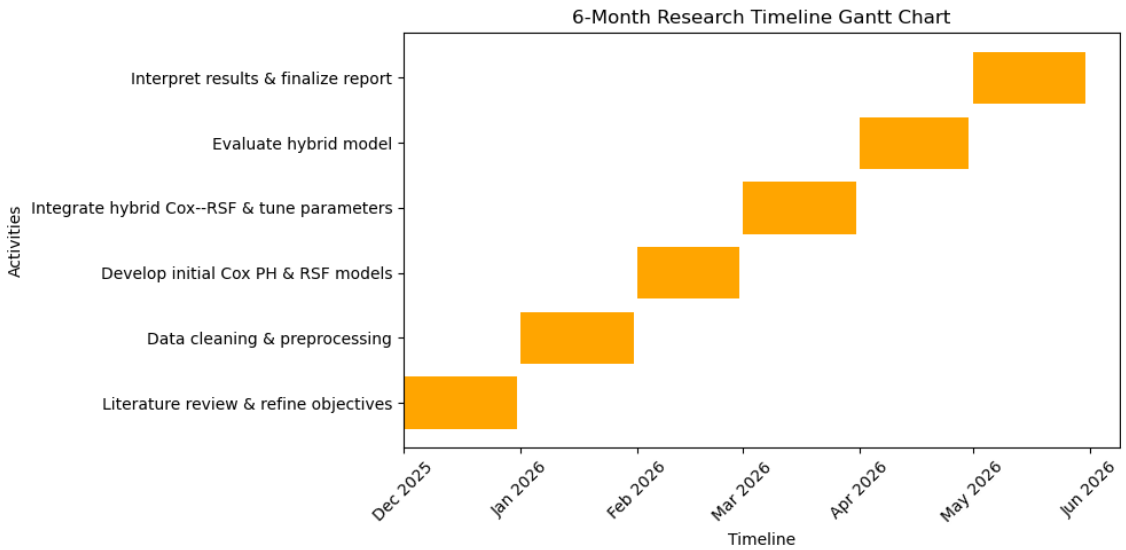
Appendix A

4.0.1 Timeline

The proposed study will be conducted over a period of six (6) months. The activities are structured to ensure systematic progress from literature review to final report submission.

Table 4.1: Proposed Timeline

Month	Activities
1	Conduct comprehensive literature review; refine research questions and objectives; obtain dataset(s) from insurers or secondary sources.
2	Perform data cleaning and preprocessing, including handling missing values, encoding categorical variables, and defining censoring indicators.
3	Develop initial models: Cox Proportional Hazards for feature selection and baseline Random Survival Forest implementation.
4	Integrate hybrid Cox–RSF model through weighted blending; perform parameter tuning using cross-validation.
5	Evaluate the hybrid model against standalone Cox PH and RSF using C-index, Integrated Brier Score (IBS), Brier Score, and time-dependent ROC.
6	Interpret results, discuss academic and practical implications, prepare final dissertation, proofreading, and submission.



Chapter 5

Appendix B

5.1 Budget

The estimated budget for the six-month project is presented in Table 5.1.

Table 5.1: Proposed Budget

Item	**Details**	**Cost (MWK)**
Data acquisition	Access/agreements for insurance datasets	200,000
Software & computing	R / Python (open-source), cloud computing credits if needed	130,000
Stationery & printing	Drafts, report printing & binding	90,000
Internet & communication	Data bundles, online resources, supervisor/industry consultations	180,000
Workshops / consultations	Meetings with supervisor, actuarial/insurance experts	200,000
Miscellaneous / contingency (10%)	Unforeseen expenses	120,000
Total		**920,000**